

USTP IC DESIGN AND VERIFICATION
DIGITAL LOGIC DESIGN
WEEK 1

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ROLL NO: 22FA-036-CE

Exercise 1

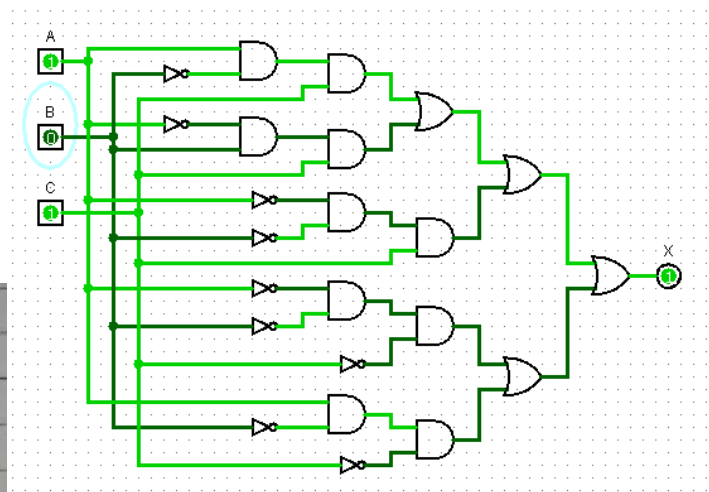
Simplify and simulate:

$$F(A, B, C) = A\bar{B}C + \bar{A}BC + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C}$$

Handwritten Karnaugh Map for Exercise 1:

A \ B \ C	00	01	11	10
0	1	1	1	
1	1	1		

Simplified expression: $B + \bar{A}C$



Exercise 2

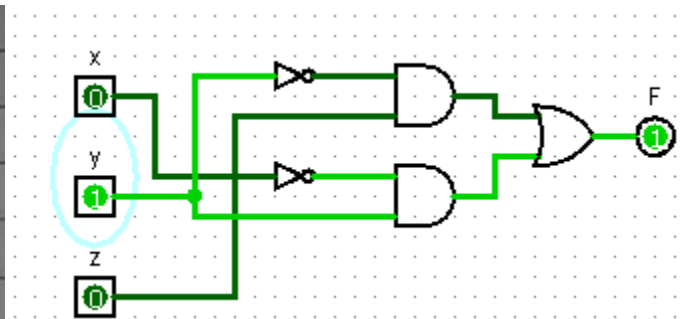
Simplify and simulate:

$$F(x, y, z) = \sum m(1, 2, 3, 5)$$

Handwritten Karnaugh Map for Exercise 2:

x \ y \ z	00	01	11	10
0	0	1	1	1
1	1	1		

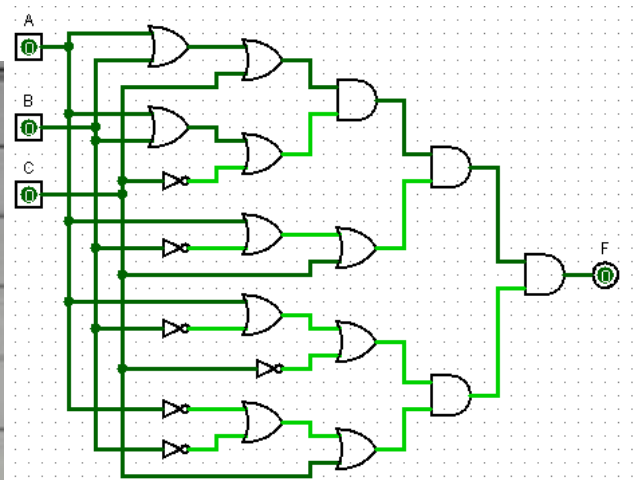
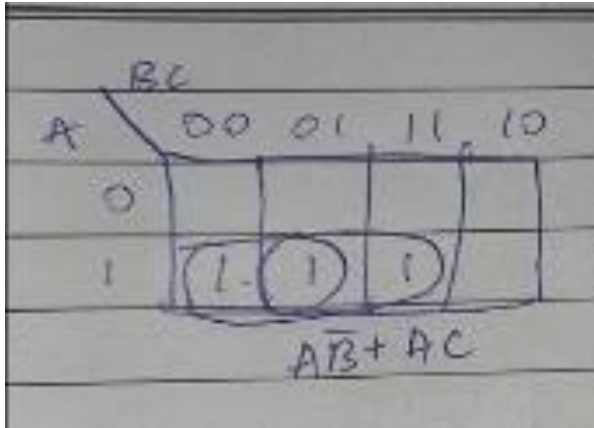
Simplified expression: $\bar{y}z + \bar{x}y$



Exercise 3

Simplify and simulate:

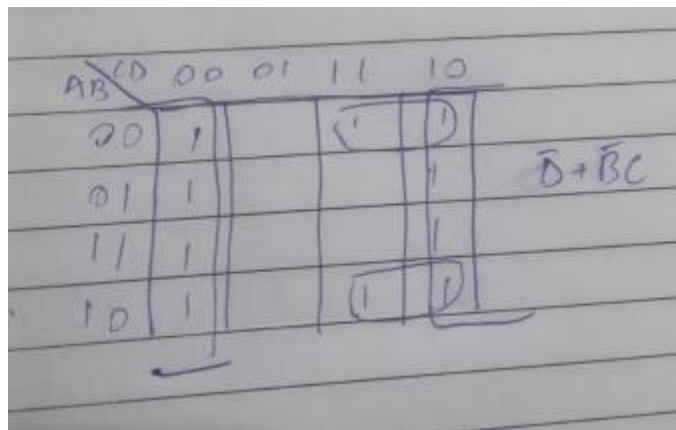
$$F(A, B, C) = (A + B + C)(A + B + \bar{C})(A + \bar{B} + C)(A + \bar{B} + \bar{C})(\bar{A} + \bar{B} + C)$$

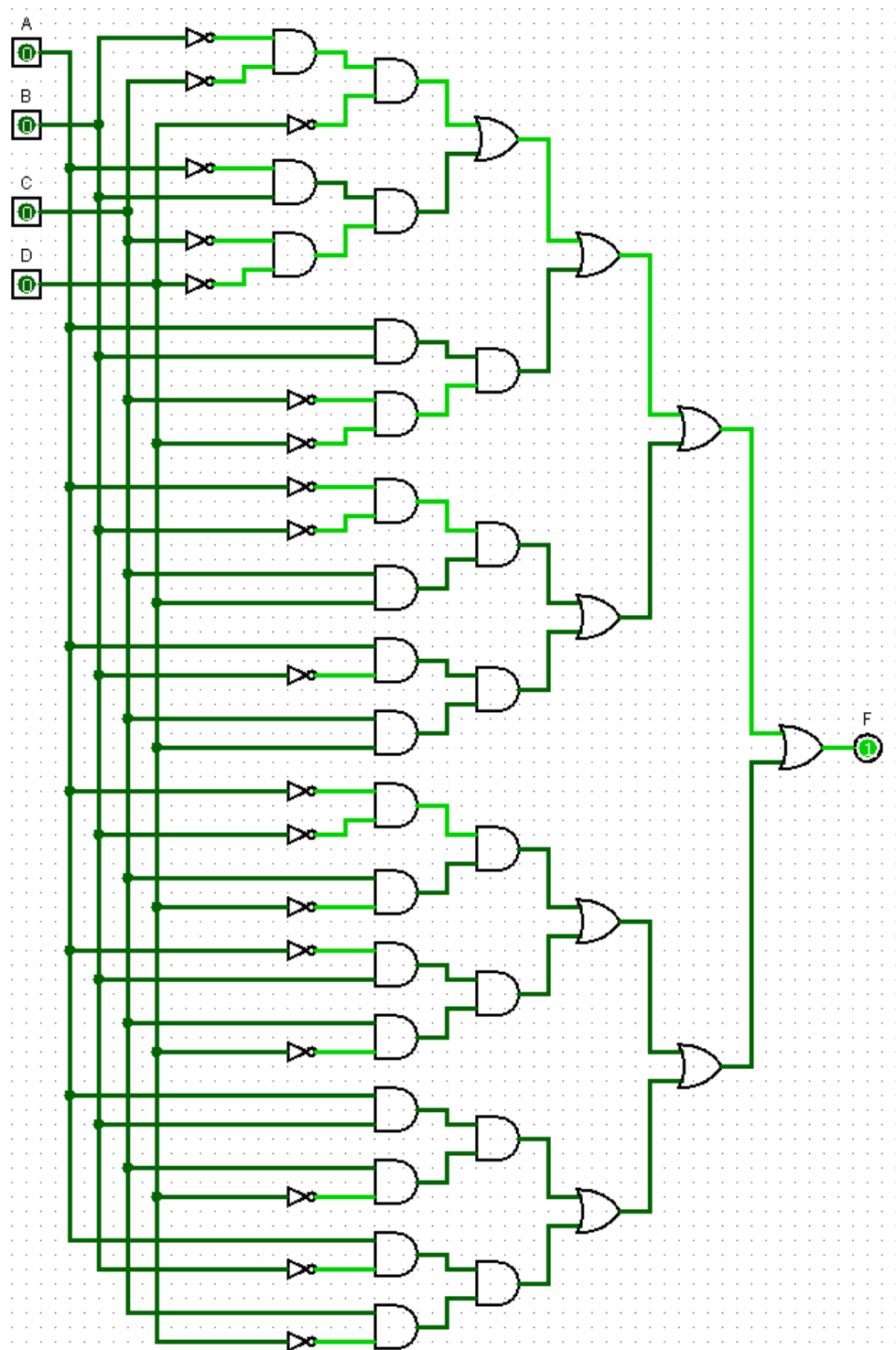


Exercise 4

Simplify and simulate:

$$F(A, B, C, D) = \bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + AB\bar{C}\bar{D} + \bar{A}\bar{B}CD + A\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}BC\bar{D} + ABC\bar{D} + A\bar{B}C\bar{D}$$

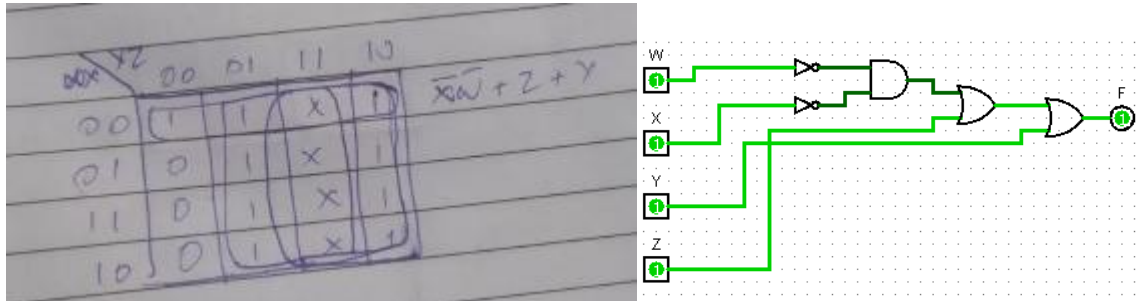




Exercise 5

Simplify and simulate:

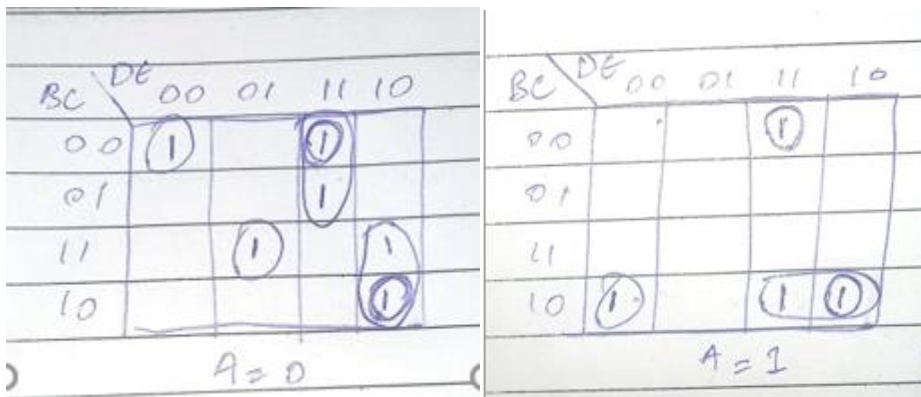
$$F(W, X, Y, Z) = \Sigma m(1, 2, 5, 6, 9, 10, 13, 14) + \Sigma d(0, 3, 7, 11, 15)$$



Exercise 6

Simplify and simulate:

$$X = A'BC'DE' + A'B'C'DE + AB'C'DE + ABC'D'E' + A'BCDE' + A'BCD'E + A'B'C'D'E' + A'B'CDE + ABC'DE' + ABC'DE$$



$$B'C'DE + BC'DE' + A'B'DE' + A'B'DE + A'BCD'E' + A'B'C'D'E' + ABC'D + AB'C'DE + ABC'DE'$$

